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MILITARY SPECIFICATION

CABINETS, FOOD SERVICE, REFRIGERATOR-WARMER, ROLL-THRU, ROLL-IN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers food service refrigerator cabinets, warmer cabinets, and combination refrigerator-warmer cabinets.

1.2 Classification. Cabinets shall be of the following types, classes, styles, inclosures, duties, and models as specified (see 6.2).

Type I - Refrigerator cabinet.

Class 1 - Single cabinet.
Class 2 - Double cabinet.
Class 3 - Triple cabinet.
Class 4 - Four section cabinet.

Inclosure 1 - Self-contained compressor/condenser unit.
Inclosure 2 - Remote compressor/condenser unit.

Duty B - Air cooled compressor/condenser unit.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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Type II - Warmer cabinet.

Class 1 - Single cabinet.
Class 2 - Double cabinet.
Class 3 - Triple cabinet.

Type III - Combination cabinets.

Class 1 - 1 warmer - 1 refrigerator.
Class 2 - 1 warmer - 2 refrigerator.
Class 3 - 2 warmer - 1 refrigerator.
Class 4 - 2 warmer - 2 refrigerator.

Inclosure 1 - Self-contained compressor/condenser unit.
Inclosure 2 - Remote compressor/condenser unit.

Duty B - Air cooled compressor/condenser unit.

Style A - Non-insulated, corrosion-resistant bottom with wall type cart guides.
Style B - Two-inch deep insulated, corrosion-resistant bottom with wheel groove cart guides.
Style C - Three-inch deep insulated bottom with wall type cart guides for installation in a recessed floor 3-3/8 inch maximum depth.
Style D - Bottomless cabinet with wall cart guides.

Model A - Roll-in.
Model B - Roll-thru.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

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MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing Of.
MIL-R-43399 - Rack, Mobile (For Roll-Thru Refrigerators and Roll-Thru Food Warmers).

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting officer or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel
A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
B 280 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS (NBS)

Standard for Devices to Permit the Opening of Household Refrigerator Doors from the Inside.

(Application for copies should be addressed to the Director, National Bureau of Standards, Washington, DC 20234.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L St., N.W., Washington, DC 20037.)

NATIONAL SANITATION FOUNDATION (NSF)

Listing of Food Service Equipment.
Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment.
Standard No. 7 - Food Service Refrigerators and Storage Freezers.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Rd., P.O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS LABORATORIES INC.

Standard No. 62 - Flexible Cord and Fixture Wire.
Standard No. 197 - Commercial Electric Cooking Appliances.
Standard No. 303 - Condensing Units Refrigeration.
Standard No. 471 - Commercial Refrigeration.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard product. The cabinets shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturers standard commercial product with any added features needed to comply with the requirements of this document. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this document shall be identifiable by all regular manufacturers or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

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3.2 Codes and standards. The cabinets shall conform to the applicable requirements of UL Standard Nos. 62, 197, 303 and 471, and NSF Standard Nos. 4 and 7. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.3).

3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial cabinets provided the completed item complies with all provisions of this document.

3.3.1 Stainless steel. Stainless steel shall conform to type 302 and 304 of ASTM A 167.

3.3.2 Structural steel. Structural steel shall conform to ASTM A 36.

3.3.3 Refrigeration tube. Tubing used for refrigerant lines shall conform to ASTM B 280. Tubing used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness not less than 0.016 inch for 3/8 inch and 1/2 inch outside diameter tube.

3.3.4 Insulation. Insulation shall be a non-moisture absorbing material; shall be homogeneous throughout; and shall have "K" factors as specified below using a 70°F mean temperature and expressed in terms of British thermal units (Btu) per hour per square foot per inch of thickness per degree F temperature difference. Loose fill type of insulation will not be acceptable. The insulation shall be of a type that is not combustible or will not support combustion, and will not release toxic fumes when exposed to a high temperature.

3.3.4.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene or polyurethane material having a manufacturers commercial rated "K" factor of not more than 0.26 for polystyrene and 0.15 for polyurethane with a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.2.2.1.

3.3.4.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturers commercial rated "K" factor of not more than 0.15, and a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.2.2.2.1.

3.3.4.3 Semirigid insulation. Semirigid insulation shall be "B" fiber or finer industrial type fiberglass with a manufacturers commercial rated "K" factor of not more than 0.260 and a density of at least 1.0 but not more than 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

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3.3.5 Refrigerant. The refrigerant shall be type 12, 22, or type 502 conforming to BB-F-1421.

3.4 Design. The cabinets shall be designed as integral free standing units in the types, classes, inclosures, duties, styles, and models specified in 1.2. All cabinets shall be designed with interior lighting, thermometers, switches, heaters, condensing units, and controls necessary for automatic operation. The refrigerator cabinets or sections shall automatically maintain temperatures at 38°F plus or minus 2°F when tested as specified in 4.4.1. The warmer cabinets or sections shall automatically maintain temperatures at 150° to 160°F, plus or minus 2°F when tested as specified in 4.4.1. The self-contained refrigerator cabinets or sections shall house the condensing unit in the top of the cabinet. The cabinet dimensions shall be as shown on figure 1. The door opening on each section shall be 65 to 66 inches by 26 to 27 inches. Each cabinet or section shall accommodate mobile racks conforming to MIL-R-43399. Unless otherwise specified (see 6.2), a mobile rack shall be furnished with each section or cabinet. All major assemblies and installed attachments shall be accessible for maintenance repair or replacement. Removable access covers and plates shall be provided as necessary to permit adjustment, service and replacement without disturbing or disassembling any major component or equipment. When specified bumpers shall be furnished on cabinets (see 3.5.8).

3.5 Construction. The cabinets shall be constructed as specified herein and as illustrated in figure 1. Figure 1, except for dimensions (width dimensions are multiples of the width of a single cabinet), is for illustrative purposes only and is not intended to preclude other cabinets otherwise meeting the requirements of this document. The style A cabinets shall be constructed having a non-insulated, corrosion-resistant bottom with wall cart guides for installation on a floor surface. The style B cabinets shall be constructed having a 2-inch deep insulated corrosion-resistant bottom with wheel groove cart guides for installation on a floor surface. The style C cabinets shall be constructed having a 3-inch deep insulated bottom with wall cart guides for installation in a recessed floor or pit a maximum depth of 3-3/8 inches. The style D cabinets shall be constructed having no bottom with wall cart guides for installation on a floor surface. The inner liner and outer shell of the cabinets shall be fabricated of stainless steel specified in 3.3.1 having a thickness of 20 gage (0.0355 inch) or greater. The space between the inner liner and outer shell shall be completely filled with the insulation specified in 3.3.4.3 when framed construction is used; when frameless construction is used the insulation specified in 3.3.4.1 or 3.3.4.2 shall be used. The interior floor for styles A, B, and C shall be constructed of steel specified in 3.3.2, having a thickness of 16 gage (0.0595 inch) or greater. Framing members, when used, shall be constructed of steel specified in 3.3.1, 3.3.2, or galvanized steel. When steel is used it shall be phosphatized and coated with a paint primer conforming to TT-P-664. Each section of the cabinets shall have full length doors with hardware. All doors shall be insulated with insulation specified in 3.3.4.1 or 3.3.4.2 and gasketed. Openings in the inner liner and outer shell for refrigerant tube, piping, and electrical fittings shall be mechanically sealed upon completion of

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the installation to prevent moisture infiltration into the insulation. The location of the openings in the cabinets shall conform to the manufacturers standard practice unless otherwise specified (see 6.2). Interior rack guides, constructed of stainless steel specified in 3.3.1 having a thickness of 0.0598 inch or greater shall be furnished in each section.

3.5.1 Electrical requirements. Unless otherwise specified (see 6.2), electrical components shall be a nominal 115-volt, single phase, 60 hertz frequency, or a 208/230 volt, single phase, 60 hertz frequency, or a combination of both. Transformers when supplied as part of the system, shall not be used to step-up or step-down the voltage for cabinet use and where specifically prohibited (see 3.6), but are permitted as part of the control system (see 3.6.2). When 50 hertz frequency is specified (see 6.2), and the cabinets are to be operated on that frequency, all motors shall be rated at that frequency. The cabinets shall be equipped with a power cord at least 6 feet long conforming to UL Standard No. 62, and terminated with a standard NEMA three prong grounding plug appropriate for the voltage and power specified. The cabinets shall be furnished completely wired and ready for connection to the electric power source.

3.5.2 Protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wet in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, by coating, by isolating, by gasketing, or by similar suitable means. Bolts, nuts, pins, screws and other fastenings and fittings of metal shall be of the same material as the materials being joined or shall be cathodic to the materials being joined.

3.5.3 Breaker strips. For refrigerator cabinets or sections, breaker strips of thermoplastic material, thermosetting plastic, natural rubber, or synthetic rubber shall be installed to join the outer shell and inner liner around exterior door openings. Joints in breaker strips shall be smooth and cleanable; edges of the breaker strips shall be eased or rounded; and the entire installation shall provide a smoothly finished appearance.

3.5.4 Condensate disposal. A condensate evaporator pan and a drain pipe shall be provided in the refrigerator sections. The condensate evaporator pan shall be stainless steel, aluminum, or glass. The drain pipe shall be brass or copper tubing or aluminum and shall include an "S" trap large enough for cleaning or be readily removable for cleaning. The condensate shall be disposed through the drain pipe to a waste drain outside the cabinet. When specified (see 6.2), condensate evaporators shall be used in lieu of the drain pipe and "S" trap.

3.5.5 Doors. The doors shall be constructed of stainless steel specified in 3.3.1. The door construction shall be frame type or frameless with low heat conductive spacers or separators, designed to prevent sagging or distortion and to provide for mounting of hardware. Gaskets shall provide an even elastic seal around the door and the method of attachment shall provide for easy replacement.

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Doors shall comply with NBS "Standard for Devices to Permit the Opening of Household Refrigerator Doors from Inside". When specified (see 6.2), door handles shall be recessed. When specified (see 6.2), self-closing doors shall be furnished.

3.5.5.1 Door latch and strike assembly. The door latch and strike assembly, when used, shall be fabricated of steel or malleable iron, copper-nickel-chromium plated; cast or forged brass, nickel-chromium plated; or high pressure die cast zinc, copper-nickel-chromium plated. Nylon contacts may be used in the latch and strike assembly. The door latch and strike assembly with handle shall be adjustable to compensate for reduction in original gasket thickness of at least 10 percent or shall be designed to automatically provide sufficient holding force to effect the gasket seal required over a range of gasket thickness reduction ranging from 0 to 10 percent. Original gasket thickness shall be measured at the latch or handle side of the door with the door closure adjusted by the manufacturer or if necessary, adjusted to effect the gasket seal required.

3.5.5.2 Hinges. Exposed hinges shall be steel or brass, copper-nickel-chromium-plated or stainless steel. Concealed or semi-concealed hinges may be steel, finished to match the exterior of the cabinets. Hinges shall be adjustable to compensate for at least a 10 percent reduction of compressible height of original gasket thickness on the hinge side of the door, except where magnetic gaskets are used, adjustment is not required.

3.5.6 Thermometer. A remote reading 2-1/2 inch dial type, flush mounted thermometer shall be installed on the front of the cabinets near the top of the door mullions. The thermometers shall have a range of 0° to 60°F for refrigerated sections, and 100° to 200°F for the warmer sections. The thermometer dial face shall be white with black marking and figures, a colored safety zone, and a recalibration feature.

3.5.7 Thermostat and pilot light. The warmer cabinets or sections shall be equipped with a thermostat calibrated from 100° to 200°F, a pilot light, and an "ON-OFF" switch located in the door mullion midway between the door handle and the top of the door mullion. The pilot light shall indicate when the heater in the warmer section is on.

3.5.8 Bumpers. When specified (see 6.2), cabinets shall be equipped with bumpers on the front and side only. Bumpers shall be fabricated from rubber like material with a durometer hardness of 75 $\pm$ 15 on the A scale when tested as specified in 4.2.2.2.2. Bumpers shall measure 3-1/2 $\pm$ 1/2 inches wide by 7/8 + 3/8 inches deep and shall be firmly assembled to the cabinet at a height of 34 $\pm$ 1/8 inches from the floor to the centerline of the 3-1/2 inch dimension. Bumpers shall be capable of being removed and replaced.

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3.6 Refrigeration system. The refrigeration system shall be hermetic type and designed for use with the refrigerant specified in 3.3.5. The refrigeration unit shall be furnished complete with a compressor/condenser unit having valves, drier, motor, and all necessary refrigeration and safety controls applicable to the refrigeration system. Provision shall be made for air distribution within the refrigerated space. All components and piping containing refrigerant shall be gas tight (see 4.4.2).

a. Inclosure 1 - Selt contained compressor/condenser unit. The self-contained refrigeration system shall be dried, evacuated, and charged at the factory with the refrigerant specified in 3.3.5 and refrigeration oil recommended by the compressor manufacturer.

b. Inclosure 2 - Remote compressor/condenser unit. The remote units shall be dehydrated and charged at the factory with a holding charge.

The refrigeration system shall operate on the voltages and frequencies specified in 3.5.1 and 3.6.2. The use of a transformer to obtain the specified voltage and frequency for the condensing unit is prohibited.

3.6.1 Evaporator assembly. The evaporator assembly for the inclosure 1 cabinets shall be either top mounted, located within a top mounted housing, or mullion type, located between the door mullions of the refrigerator section. The evaporator assembly for the inclosure 2 cabinets shall be mullion type, located within the door mullions of the refrigerator section. Each evaporator assembly shall be furnished complete with evaporator fan and fan motor, and all accessories, instruments and controls necessary to provide and automatically maintain the refrigerator temperature specified in 3.4. All components of the evaporator assembly shall be contained within a protecting shroud or housing.

3.6.2 Refrigerant controls. Controls shall be provided to automatically maintain the specified cabinet temperatures. Motor cutout switches shall open the motor circuit when the discharge pressure becomes excessively high or when the suction pressure becomes excessively low. The automatic control equipment shall be positive in action, and shall be adjustable through the temperature or pressure range required for the temperatures specified in 3.4. Controls shall be designed for operation on a nominal 208/230 volt, single phase, 60 hertz system, or on a 115-volt, single phase, 60 hertz system (see 3.5.1). When a 50 hertz system is specified (see 6.2) controls shall comply with the operational and electrical requirements. Controls rated at 115-volt, single phase, 60 hertz system may be used with a 50 hertz frequency system provided that the power is obtained from either a hot line and neutral, or through a transformer and that the overall performance of the system is not affected. Relays and transformers or any other component necessary for a complete control system, shall be furnished as an integral part of the cabinet.

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3.6.3 Piping. All refrigerant piping shall be the copper tubing specified in 3.3.3. The tubing shall be arranged so as not to interfere with the servicing or removal of the compressor/condenser unit. Fittings shall be of wrought or forged copper. The refrigeration unit design shall prevent condensation on suction lines within the machine compartment. The run of suction line exposed to room air of inclosures 1 and 2 units including the section in the machine compartment of inclosure 2 units shall be insulated with flexible preformed 3/8-inch plastic foam material. Linear and butt ends and terminal ends shall be sealed with adhesive recommended for this application by the same manufacturer supplying the insulation. Where liquid and suction lines are fastened together, the insulation shall be installed to cover both lines. All tubing connections shall be soldered, except flared connections shall be provided on expansion valve inlet and outlet connections, dehydrator inlet and outlet connections, receiver outlet valve, and any other locations normal in manufacturer's standard design. Where tubing of different materials are joined, a shrink-fit plastic sleeve or other method shall be used to positively prevent wetting of the surfaces for at least 1-inch to each side of the joint.

3.6.4 Electromagnetic compatibility. When specified (see 6.2), the cabinets shall be designed and equipped for electromagnetic compatibility in accordance with class C3, UM05, group I of MIL-STD-461 (see 4.4.3).

3.7 Finish. The cabinets shall have a number 5 finish or better on the exterior fronts, sides, top, doors, and backs; except the roll-in type backs and all exterior floors may be galvanized steel. The interior shall be stainless steel having a number 4 finish or better.

3.8 Data-name plates. The cabinets shall be furnished with a data-name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the cabinets by rivets, screws or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (See 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the cabinet.

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3.9 Instruction plate. Instruction plates shall be made of corrosion-resisting metal and shall be attached to the cabinets by either rivets, screws or welding. The instruction plates shall have the wiring diagram and refrigerator cycle diagram. The plates shall also contain instructions for start-up, operation, shutdown, preventative maintenance, and the type of refrigerant as supplied in the system. In addition, information as required by UL Standard No. 471 shall also appear on the plate.

3.10 Workmanship. The finished cabinets shall be complete and clean, and free of fabrication marks, scratches, dents, breaks, sharp edges, sharp corners, deformities, burrs, slivers, or visible evidence of rust.

3.10.1 Rivets. Rivets, when used, shall be tight and the rivet heads and peened ends shall be round and concentric and shall be in full contact with the joined metal surfaces. There shall be no empty rivet holes.

3.10.2 Welds. Welds shall be clean, continuous, sound, free of flux, pits, buildups, undercuts, and shall be ground so as to blend with adjacent surfaces. Spot welds when used shall be sound and free of discoloration, splatter and burn throughs.

3.10.3 Threaded fasteners. Threaded fasteners shall be drawn tight. Lockwashers shall be used on all nuts and bolts. There shall be no crossed or stripped threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 In-process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, corrections shall be made to the operation and items affected.

a. Installation of insulation (see 3.5).
b. Sealing of openings in cabinet shells against moisture infiltration (see 3.5).
c. Steel surfaces are primed (see 3.5).
d. Protection of joined dissimilar metals against corrosion (see 3.5.2).
e. Internal construction of doors (see 3.5.5).
f. Installation of piping and accessories (see 3.6.3).

4.2.2.2 In-process testing.

4.2.2.2.1 Density of insulation, (when pre-expanded or foamed-in-place insulation is used. Prior to the beginning of each day's production of cabinets, one sample of the insulation to be used in that day's production shall be tested as specified in 4.4.4 for conformance to the applicable density requirements as specified in 3.3.4.1 and 3.3.4.2.

4.2.2.2.2 Bumper material hardness. The bumper material shall be tested for hardness in accordance with Method 3201 of FED-STD-601. The sample shall be one bumper. The lot shall consist of all bumpers offered for acceptance at one time. Failure of the sample to comply with the hardness requirement in 3.5.8 shall be cause for rejection of the lot.

4.2.3 End item examination.

4.2.3.1 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of cabinets. The sample unit shall be one completely fabricated cabinet. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers standard product except for changes necessary to comply with document requirements	X	
Finish	Not as specified	X	
	Rough and sharp edges, burrs, or slivers	X	
Design, construction and workmanship, general	Not self supporting	X	
	Access panels, covers, and plates not provided for maintenance	X	
	Component not as specified	X	
Outer shell and inner liner	Not welded construction	X	
	Openings for tubing, piping and electrical connections not mechanically sealed	X	
Breaker strip	Not installed as specified	X	
	Edges not eased or rounded		X
	Joints not sealed	X	
Doors	Number of doors more or less than specified	X	
	Gasket does not provide even seal	X	
	Gasket not easily replaced		X
	Not self-closing (when required)	X	
	Handles not recessed (when required)	X	
Evaporator	Not type specified	X	
	Fan or fan motor missing	X	
	Protecting shroud or housing missing	X	
Bumpers	Missing, not type specified	X	
	Not removable		X
Thermometer	Not type or range specified	X	
	Not installed where specified		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Thermostat and pilot lights	Not located as specified		X
Electrical	Cord and plug do not provide grounding (when required)	X	
	Switch location not as specified		X
	Switch not clearly marked	X	
Bolts, nuts, and other threaded fasteners	Broken, stripped, missing, fractured		X
	Not drawn tight, or not secure		X
Workmanship	Rivets not tight, head and end not round and concentric, not in full contact with joined surface		X
	Sheet metal kinked or bent		X
	Shearing and chipping, ragged, not neatly finished		X
	Bolt or rivet holes not accurately drilled or punched		X
	Accessories and piping not securely fastened or braced		X
Welding	Not free from rust scale, paint, grease, and other foreign matter	X	
	Scale and flux deposits not removed		X
	Weld not smooth and free of cracks, burnholes, or undercuts		X
	Weld not smooth due to incomplete fusion		X
Data-name plates	Omitted or not as specified		X
	Information incomplete or not legible		X
	Not located so as to readily visible to operator		X
Instruction plate	Content of instruction plate: Missing, incomplete, illegible	X	

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4.2.3.2 End item dimensional examination. The end item shall be examined for the dimensional requirements specified in 3.5 and figure 1. The lot size shall be expressed in units of cabinets. The sample unit shall be one complete cabinet. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 End item testing.

4.2.4.1 Refrigerant leakage testing. Each refrigerator cabinet or section, shall be tested as specified in 4.4.2. Any end item exhibiting one or more test failures shall be rejected.

4.2.4.1.1 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.1 and in 4.4.3 (when applicable). Any test failure shall be cause for rejection of the lot.

4.2.5 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-12323.

4.3 Certification compliance examination. Certifications, certified test reports or listing marks for standards, submittted in accordance with 3.2 shall be examined and validated as proof of compliance. Prior to Government approval of first shipment the contractor shall submit to the contracting officer or his authorized representative for approval certification that the cabinets he proposes to furnish meet the applicable requirements of UL and NSF as follows:

4.3.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the cabinets offered have been tested and conform to UL Standard Nos. 62, 197, 303 and 471.

4.3.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished cabinet, or

(2) A certification for the cabinet issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report acceptable to the contracting officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating the cabinets have been tested and conform to the specified NSF Standards.

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4.4 Methods of inspection.

4.4.1 Performance test. The performance test shall be conducted in an ambient temperature of 110°F. The refrigerated sections shall be precooled to a temperature of 38°F, plus or minus 2°F, while the food warmer sections are maintained at 150° to 160°F. The precooling time shall be not less than 5 hours before the test run. The control on the refrigerator cabinets shall be set at 38°F and the controls shall be set at 155°F for the warmer cabinets and the test run for a period of 8 hours. During this time observation shall be made that all components function as intended and that the compressor on the refrigerator cabinets does not cycle or cut out on overload protectors. Inability of the cabinets to maintain the temperatures required in 3.4, failure of components to function as intended, or compressor failure shall constitute failure of this test.

4.4.2 Refrigeration leak test. The refrigerator cabinets shall be refrigerant leak tested to determine conformance with 3.6. The test for leakage shall be made with a halide leak detector or electronic leak detector. Any refrigerant leak shall constitute failure of this test.

4.4.3 Electromagnetic compatibility. When electromagnetic compatibility is required, the production unit shall be tested by the contractor in accordance with test methods of MIL-STD-462 for UM05 requirements. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and requirements meet MIL-STD-461.

4.4.4 Insulation density tests.

4.4.4.1 Free rise method. The sample shall be a 6-inch cube cut from the center of a minimum 8-inch free foam cube of the insulation. The 6-inch cube shall be weighed and the density shall be calculated as follows:

$$\text{Density (lbs./ft.}^3\text{)} = \text{Weight of 6-inch cube (lbs.)} \times 8$$

Any non-conformance with the above shall constitute failure of this test.

4.4.4.2 Foamed-in-place method. The weight and interior volume of a refrigerator-warmer cabinet panel without foam insulation shall be determined. The panel shall then be foamed and weighed. The foam density shall then be calculated as follows:

$$\text{Density (lbs./ft.}^3\text{)} = \frac{\text{Panel weight (lbs.) after foaming} - \text{Panel weight (lbs.) before foaming}}{\text{Panel interior volume}}$$

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Any non-conformance with the above shall constitute failure of this test.

5. PACKAGING

5.1 Levels A, B, and C. Each refrigerator or food warmer cabinet or combination thereof shall be preserved, packaged, packed and marked for each level of protection in accordance with MIL-R-12323 (see 6.2).

6. NOTES

6.1 Intended use. The refrigerator sections are intended for holding food at 38°F prior to serving. The food warmer sections are intended to hold food at warming temperatures from 150° to 160°F prior to serving.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, class, style, inclosure, duty, or model cabinet required (see 1.2).
c. When mobile racks are not required (see 3.4).
d. Location in outer shell for refrigerant pipe and electrical connections when other than manufacturer's standard (see 3.5).
e. When electrical characteristics, other than those specified are required, the contracting officer shall provide voltage, phase, cycle and other necessary requirements to facilitate operation and installation (see 3.5.1).
f. When condensate evaporator is required (see 3.5.4).
g. When recessed door handles are required (see 3.5.5).
h. When self-closing doors are required (see 3.5.5).
i. When bumpers are required on cabinets (see 3.5.8).
j. When electromagnetic compatibility is required (see 3.6.4).
k. Government approved manual number to be included on data-name plate (see 3.8).
l. Selection of applicable levels of preservation and packaging and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force - 99

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Navy - MC
Air Force - 84
DLA - GS

User activity:

Navy - SA

Preparing activity:

Army - GL

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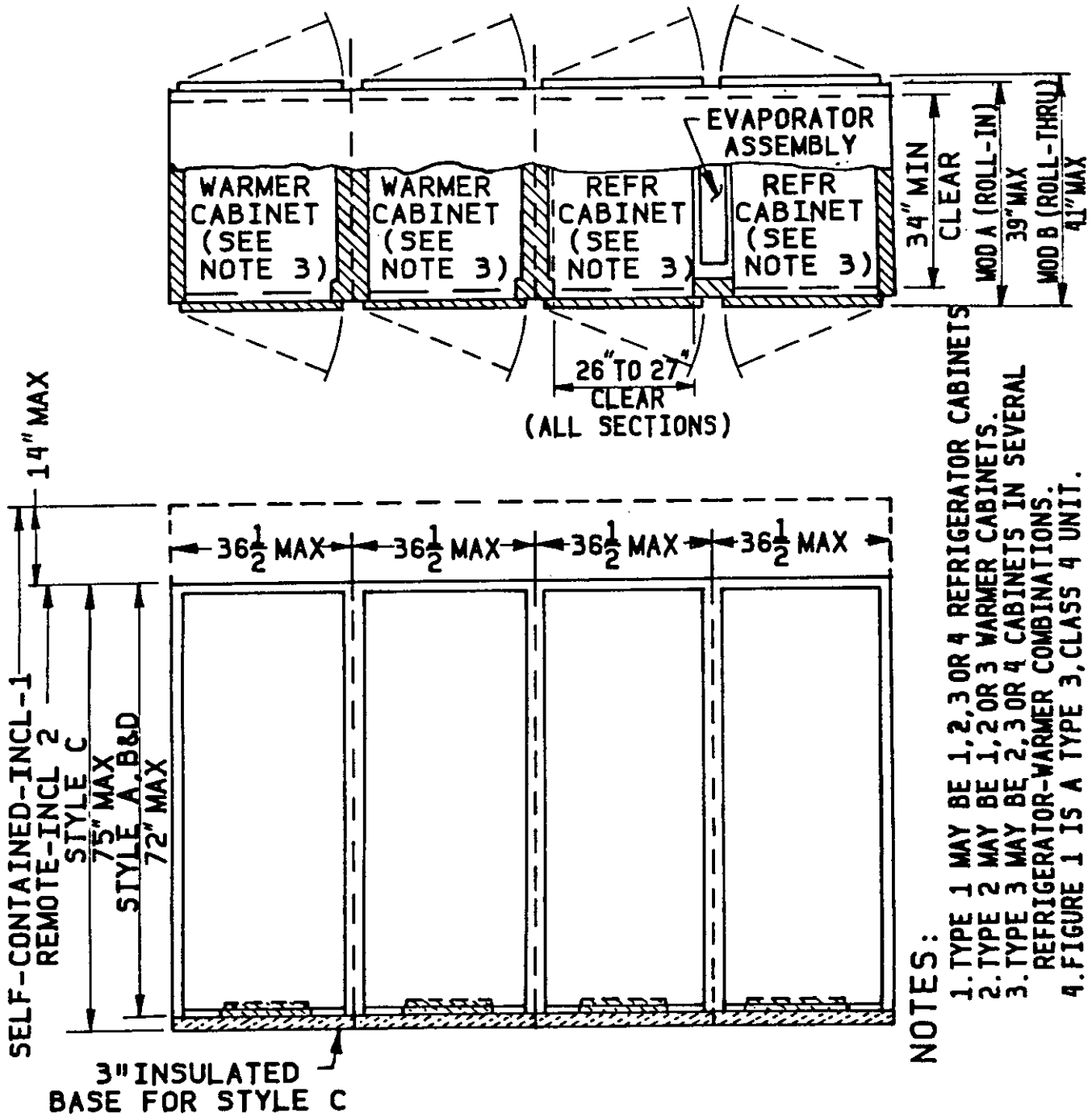


FIGURE 1 CABINETS, FOOD SERVICE, WARMER-REFRIGERATOR, ROLL-THRU, ROLL-IN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43427C	Cabinets, Food Service, Refrigerator-Warmer, Roll-Thru, Roll-In

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK/OP1, 29 Nov 84

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